Analysis of Factors Associated with Poor Response to Nutritional Treatment in Severely Malnourished Children Aged 6–59 Months With Edema and With Complications Treated (Phase T).

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1. Methodology

Descriptive analyses were conducted to characterize the cohort and the non-response to the nutritional treatment. Continuous variables were summarized using medians and interquartile ranges (25th - 75th percentile) due to their non-normal distribution, whereas categorical variables were summarized as counts and percentages (%).

Differences between groups (success and failure) were assessed using the Wilcoxon rank-sum test or Kruskal–Wallis test for continuous variables and Pearson's chi-square or Fisher's exact test for categorical variables, as appropriate.

Univariable logistic regression analyses were performed to estimate crude associations between independent variables and the dependent variable. Variables considered clinically relevant or statistically associated with the non-response to the nutritional treatment at $\underline{p} < 0.20$ were included in a multivariable logistic regression.

Prior to model building, pairwise correlations among independent variables were examined to identify highly correlated predictors and assess potential multicollinearity. Pearson correlation coefficients were computed for continuous variables (or Spearman's rank correlations for non-normally distributed variables). Variables exhibiting strong correlations ($|r| \geq 0.70$) were carefully evaluated to avoid redundancy. Variance Inflation Factors (VIFs) were subsequently calculated in the multivariable model, with VIF values > 5 indicating potential multicolliniarity.

Model selection was guided by likelihood ratio tests and the principle of parsimony. A stepwise selection procedure was used to identify the most parsimonious model (based on the Akaike Information Criterion (AIC)) by retaining variables that improved model fit while excluding predictors that contributed little additional information.

Adjusted odds ratios (aORs) with 95% confidence intervals (CIs) were reported. Statistical significance in the final model was defined as a two-sided $\underline{p} < 0.05$. A variable was significantly associated with the risk of nutritional treatment failure if: i) the odds ratio (OR) was > 1 with a lower limit of the confidence interval (CI) > 1; or ii) OR<1 with upper CI limit <1; and p-value was <0.05.

All analyses were conducted using R (version 4.3.0).

2. Study population

Insert comments here.

3. Descriptive Characteristics

Insert comments here.

4. Bivariate Analysis

Table 1: Association between explanatory variables and outcome at phase 1.

	Non-response to treatment in Phase				
Characteristic	**FALSE** N = 125[1]	**TRUE** N = 50[1]	**Overall** N = 175[1]	**p-value**[23]	**q-value**[4]
Age (Month)				0.389	0.472
Median (Q1 ; Q3)	26 (19.0 ; 32)	25 (18.0 ; 30)	26 (18.0 ; 32)		
Min ; Max	7 ; 59.0	8 ; 48.0	7 ; 59.0		
Age group				0.811	0.862
[25;59]	65 (52.0%)	25 (50.0%)	90 (51.4%)		
[6;25[	60 (48.0%)	25 (50.0%)	85 (48.6%)		
Gender				**0.039***	0.148
Male	74 (59.2%)	21 (42.0%)	95 (54.3%)		
Female	51 (40.8%)	29 (58.0%)	80 (45.7%)		
Locality				0.094	0.216
Urban	23 (18.9%)	15 (30.6%)	38 (22.2%)		
Rural	99 (81.1%)	34 (69.4%)	133 (77.8%)		
Missing values	3	1	4		
Vaccination status				0.191	0.324
Up to date	39 (81.3%)	11 (64.7%)	50 (76.9%)		
Out of date	9 (18.8%)	6 (35.3%)	15 (23.1%)		
Missing values	77	33	110		
Breastfeeding status	64 (88.9%)	27 (79.4%)	91 (85.8%)	0.235	0.381
Missing values	53	16	69		
Type admission				0.984	0.984
New admission	108 (89.3%)	42 (89.4%)	150 (89.3%)		
Readmission	13 (10.7%)	5 (10.6%)	18 (10.7%)		
Missing values	4	3	7		
Morbidity burden				**<0.001*****	0.001

Characteristic	Non-response to treatment in Phase				
	FALSE N = 125[1]	**TRUE** N = 50[1]	**Overall** N = 175[1]	**p-value**[23]	**q-value**[4]
Median (Q1 ; Q3)	1 (1.0 ; 2)	2 (1.0 ; 3)	2 (1.0 ; 3)		
Min ; Max	0 ; 4.0	0 ; 5.0	0 ; 5.0		
Shock	0 (0.0%)	1 (2.0%)	1 (0.6%)	0.286	0.389
Gastroenteritis	56 (44.8%)	38 (76.0%)	94 (53.7%)	**<0.001*** **	0.002
Pneumonia	53 (42.4%)	27 (54.0%)	80 (45.7%)	0.164	0.294
Tuberculosis	0 (0.0%)	1 (2.0%)	1 (0.6%)	0.286	0.389
Digestive candidiasis	13 (10.4%)	12 (24.0%)	25 (14.3%)	**0.020***	0.111
HIV	0 (0.0%)	1 (2.0%)	1 (0.6%)	0.286	0.389
Severe malaria	43 (34.4%)	16 (32.0%)	59 (33.7%)	0.762	0.851
Neurological malaria	2 (1.6%)	2 (4.0%)	4 (2.3%)	0.322	0.406
Severe anemia	25 (20.0%)	17 (34.0%)	42 (24.0%)	0.050	0.155
Other anemia	25 (20.0%)	17 (34.0%)	42 (24.0%)	0.050	0.155
Persistent edema in day11 (PT)	0 (0.0%)	16 (32.0%)	16 (9.1%)	**<0.001*** **	<0.001
Phase length (PT)				**0.023***	0.111
Median (Q1 ; Q3)	3 (2.0 ; 4)	2 (1.0 ; 3)	2 (2.0 ; 4)		
Min ; Max	1 ; 19.0	1 ; 8.0	1 ; 19.0		
Phase T to Phase 1 transition	0 (0.0%)	38 (76.0%)	38 (21.7%)	**<0.001*** **	<0.001
Nutritional treatment (PT)				0.922	0.950
F75+RUTF/F75F+RUTF	34 (27.2%)	14 (28.0%)	48 (27.4%)		
F100	56 (44.8%)	21 (42.0%)	77 (44.0%)		
F75/F75F	9 (7.2%)	5 (10.0%)	14 (8.0%)		
Others	26 (20.8%)	10 (20.0%)	36 (20.6%)		
F75+RUTF to F100 transition (PT)	0 (0.0%)	0 (0.0%)	0 (0.0%)		
F75 frequency below target (PT)				0.089	0.216
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F75 frequency above target (PT)					
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 0.0	0 ; 0.0	0 ; 0.0		
F75 quantity below target (PT)				**0.012***	0.081
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 1)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F75 quantity above target (PT)				0.305	0.399

Characteristic	Non-response to treatment in Phase				
	FALSE N = 125[1]	**TRUE** N = 50[1]	**Overall** N = 175[1]	**p-value**[23]	**q-value**[4]
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F75 adherence below target (PT)				**0.032***	0.135
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 1)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F75 adherence above target (PT)				0.086	0.216
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F100 frequency below target (PT)					
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 0.0	0 ; 0.0	0 ; 0.0		
F100 frequency above target (PT)				0.776	0.851
Median (Q1 ; Q3)	1 (0.0 ; 1)	0 (0.0 ; 1)	0 (0.0 ; 1)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F100 quantity below target (PT)				0.700	0.821
Median (Q1 ; Q3)	0 (0.0 ; 1)	0 (0.0 ; 1)	0 (0.0 ; 1)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F100 quantity above target (PT)				0.117	0.226
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 0.0	0 ; 1.0	0 ; 1.0		
F100 adherence below target (PT)				0.280	0.389
Median (Q1 ; Q3)	0 (0.0 ; 1)	0 (0.0 ; 1)	0 (0.0 ; 1)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F100 adherence above target (PT)				0.117	0.226
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 0.0	0 ; 1.0	0 ; 1.0		
Death (PT)	0 (0.0%)	1 (12.5%)	1 (1.2%)	0.095	0.216
Missing values	49	42	91		
Type of discharge (PT)				0.119	0.226
Recovered/Transferred	68 (89.5%)	7 (87.5%)	75 (89.3%)		
Absconded	8 (10.5%)	0 (0.0%)	8 (9.5%)		
Death	0 (0.0%)	1 (12.5%)	1 (1.2%)		
Missing values	49	42	91		

Characteristic	Non-response to treatment in Phase				
	FALSE N = 125[1]	**TRUE** N = 50[1]	**Overall** N = 175[1]	**p-value**[23]	**q-value**[4]

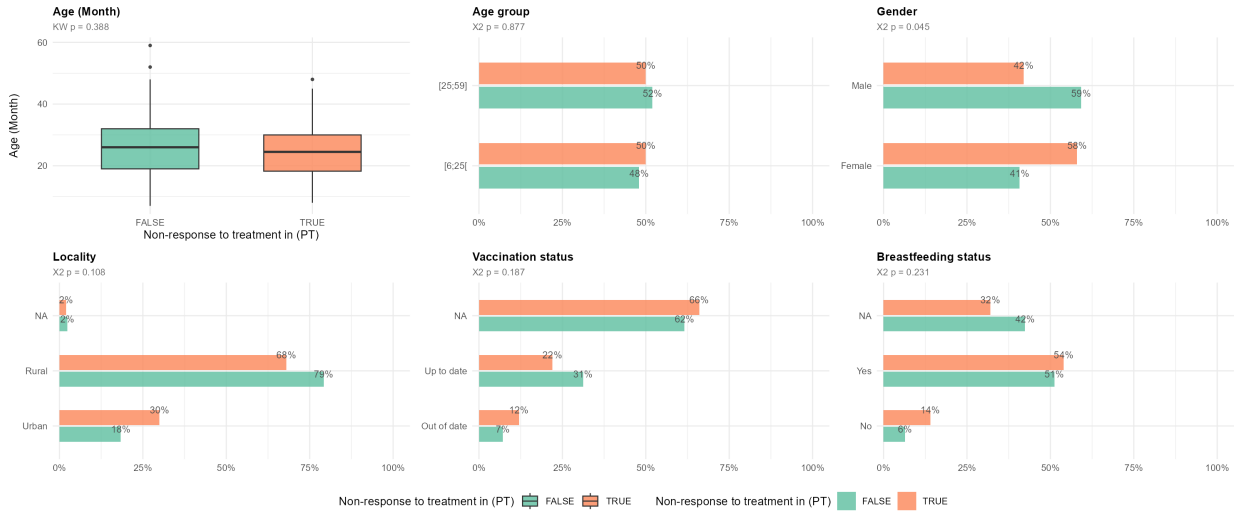
[1] n (%)

[2] Wilcoxon rank sum test; Pearson's Chi-squared test; Fisher's exact test; NA

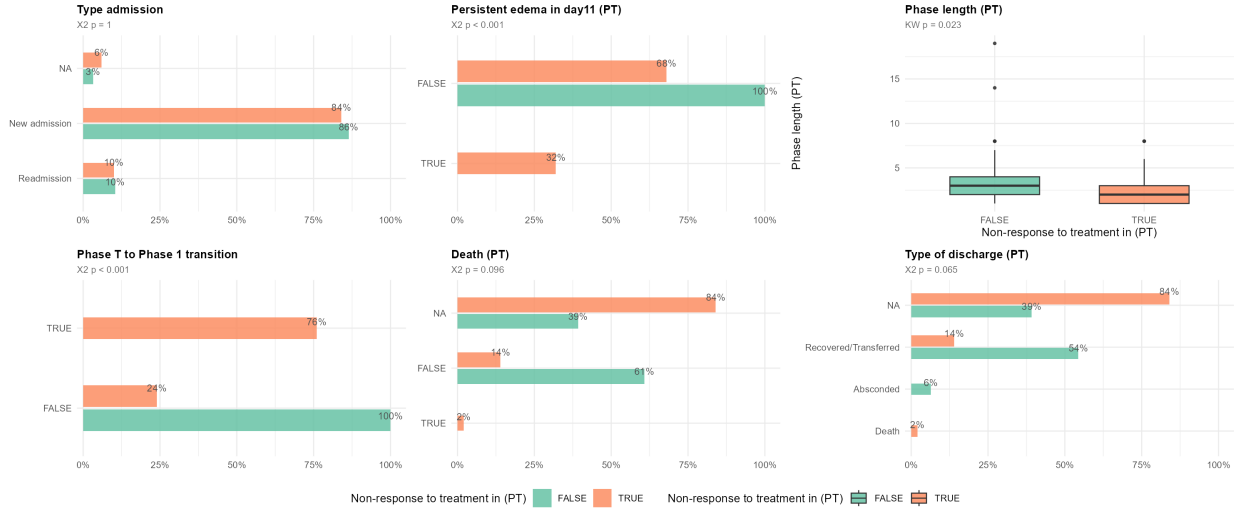
[3] *p<0.05; **p<0.01; ***p<0.001

[4] False discovery rate correction for multiple testing

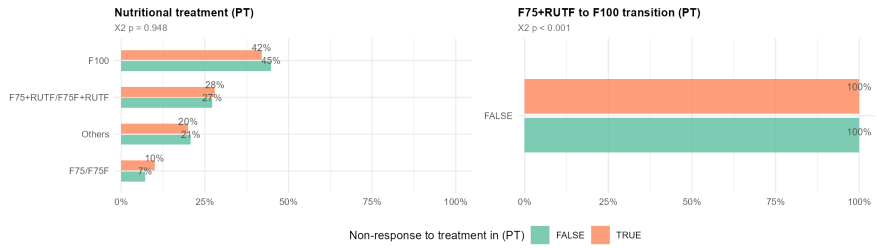
Sociodemographic



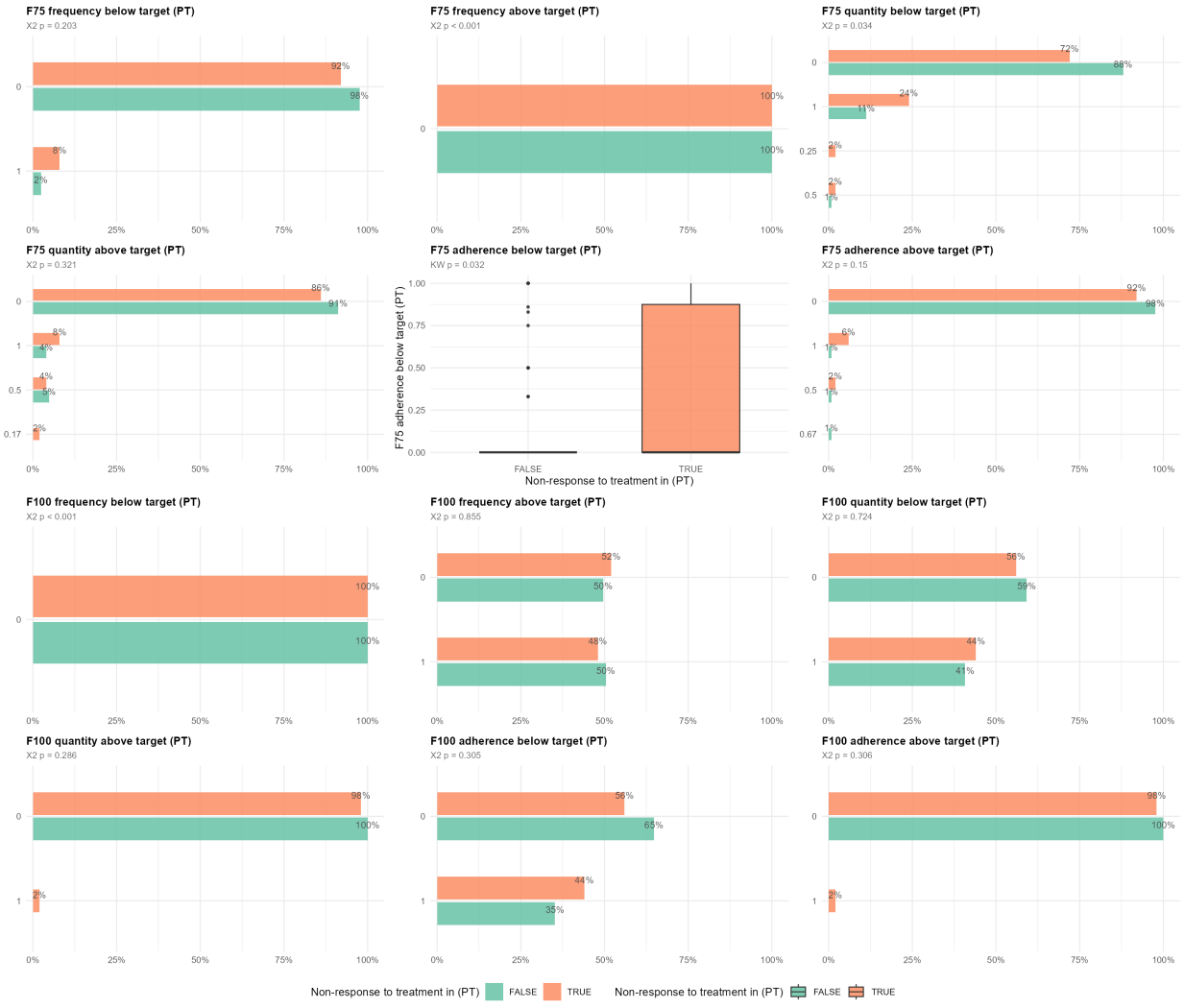
Clinical Admission and Severity



Nutritional Treatment

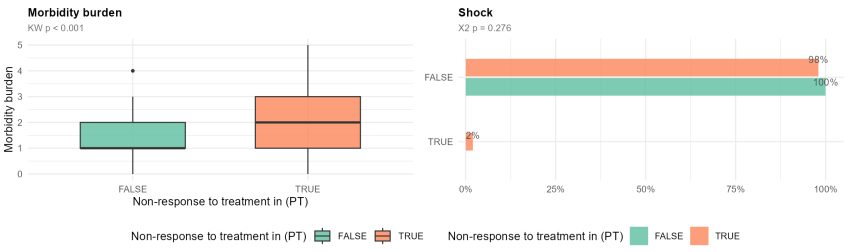


Feeding Compliance

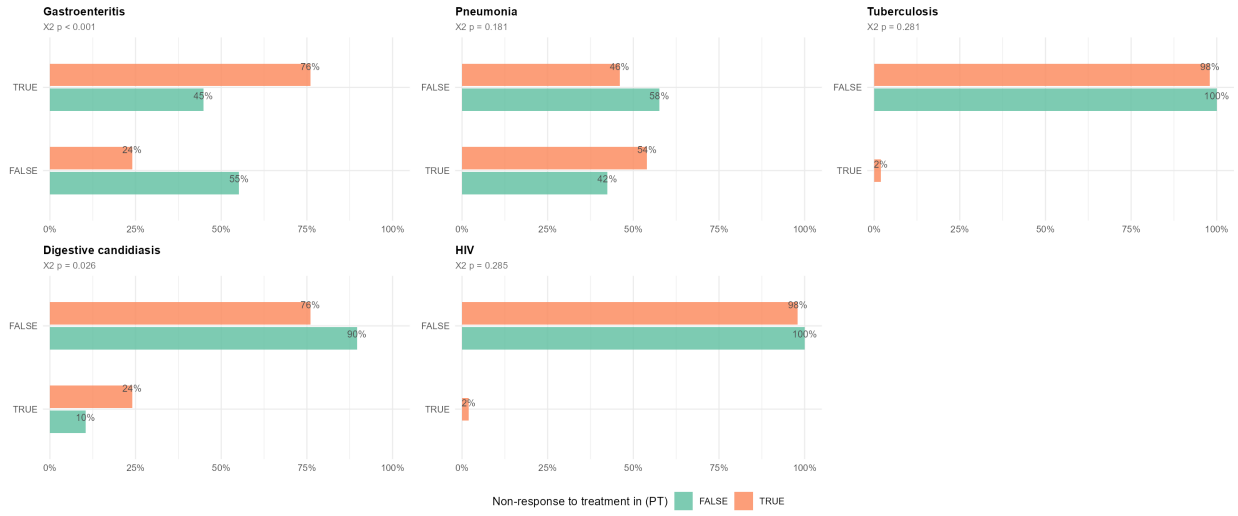


Morbidity and Comorbidities

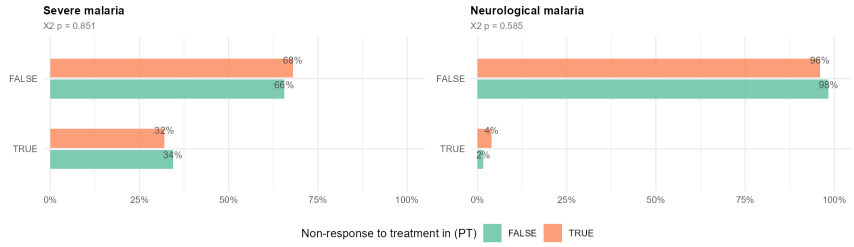
General



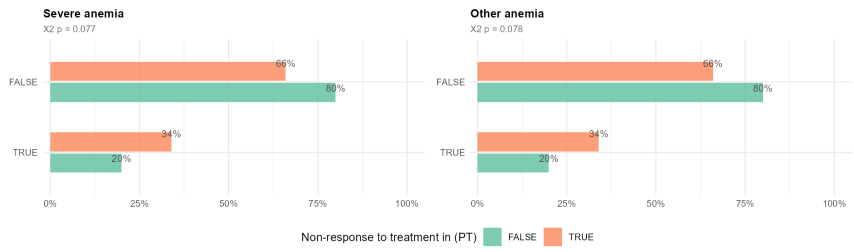
Infections and Parasitic



Malaria



Anaemia



Insert comments here.

Table 2: Results of the univariate logistic regression analysis between explanatory variables and outcome.

Predictor	Event N	OR	95% CI	p-value
Gastroenteritis				**<0.001**
FALSE	12	—	—	
TRUE	38	3.90	1.91, 8.45	
Digestive candidiasis				**0.026**
FALSE	38	—	—	
TRUE	12	2.72	1.13, 6.51	
Gender				**0.039**
Male	21	—	—	
Female	29	2.00	1.04, 3.94	
Severe anemia				0.055
FALSE	33	—	—	
TRUE	17	2.06	0.98, 4.27	
F75 adherence below target (PT)				0.057
0	34	—	—	
]0;100]	16	2.09	0.98, 4.40	
F75 adherence above target (PT)				0.11
0	46	—	—	
]0;100]	4	3.54	0.75, 18.5	
Pneumonia				0.2
FALSE	23	—	—	
TRUE	27	1.59	0.83, 3.10	
Severe malaria				0.8
FALSE	34	—	—	
TRUE	16	0.90	0.44, 1.79	
Age group				0.8
[25;59]	25	—	—	
[6;25[	25	1.08	0.56, 2.09	

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Insert comments here.

5. Multivariate logistic regression analyses

Table 3 : Results of the generalize logistic regression analysis at Phase T.

Predictor	**N**	**Event N**	**OR**[1]	**95% CI**	**p-value**	**q-value**[2]	**VIF**
Gastroenteritis	175		***		<0.001	**0.001**	1.0
FALSE		12		—			
TRUE		38	3.64	1.75, 7.98			
Gender	175		*		0.050	**0.074**	1.0
Male		21		—			
Female		29	2.00	1.00, 4.07			
Severe anemia	175				0.2	0.2	1.0
FALSE		33		—			
TRUE		17	1.77	0.80, 3.84			

[1]*p<0.05; **p<0.01; ***p<0.001

[2]False discovery rate correction for multiple testing

Abbreviations: CI = Confidence Interval, OR = Odds Ratio, VIF = Variance Inflation Factor

Insert comments here.